Kelly Capital Growth Investment Criterion — Full-Book ANSWER KEY

MacLean, Thorp & Ziemba (eds.) — World Scientific, 2011

I. Kelly's Original Formula (Chs. 1–6)

1. The Kelly criterion maximizes the expected growth rate of the logarithm of wealth over repeated bets.
2. For a coin toss with win probability p and loss probability q, the optimal fraction is f* = p − q.
3. Kelly (1956) showed that a gambler following f* will eventually outperform / surpass any opponent using a fixed fraction strategy.
4. The Kelly bettor's wealth grows at the geometric rate $g_\infty(f) = r + f(m-r) - \sigma^2 f^2/2$, where σ^2 is the variance of the asset return.
5. In the continuous-time setting, the unconstrained optimal fraction is f* = (m−r) / σ^2.

II. Breiman & Thorp — Theoretical Foundations (Chs. 7–12)

6. Breiman (1961) proved that Kelly betting minimizes the expected time to reach any fixed wealth goal.
7. Thorp and Kassouf (1967) applied Kelly to warrant / options markets, showing systematic edge over the house.
8. The capital growth log-optimal portfolio $F^* = C^{-1}[M - R]$, where C^{-1} is the inverse of the covariance matrix.
9. For N independent bets with log-normal returns, the Kelly fraction vector satisfies $F^* = \Sigma^{-1} / C^{-1}$ (M − R).
10. Thorp (1971) demonstrated that a portfolio / scientific strategy can achieve the theoretical Kelly growth rate in practice.

III. Continuous-Time Theory (Chs. 13–18)

11. In the Merton (1969) framework, the log-optimal portfolio maximizes logarithmic utility of terminal wealth.
12. The Ornstein–Uhlenbeck (OU) process for a mean-reverting asset implies $f^* = R^{-1}c_t$, where c_t is the expected return vector.
13. For an OU process with long-run mean μ and current level X_t, the drift is $\kappa(\mu - X_t)$, where κ is the mean-reversion speed.
14. The stochastic differential equation for an OU wealth process under Kelly is $dW = W \cdot f^*(\mu_t - r)/\sigma^2 \cdot (\mu_t - r)\, dt$.
15. Under OU dynamics, the optimal fraction at time t is f* = (μ_t − r) / σ^2.

IV. Good and Bad Properties (Chs. 19–24)

16. A key advantage of Kelly is that it dominates / outperforms any essentially different strategy in the long run.
17. One criticism of Kelly is that it can require large position sizes and extreme volatility / drawdowns.
18. The 'bad property' identified by Samuelson is that Kelly maximizes geometric mean but ignores investor utility / risk aversion.
19. MacLean, Thorp & Ziemba showed that even a half-Kelly bettor retains 75 % of the growth rate while cutting variance by 75%.
20. The 'overbetting' danger means that using f > f* reduces long-run growth rate and can lead to ruin.

V. Fractional Kelly & Utility (Chs. 25–30)

21. A fractional Kelly strategy uses fraction λf*, where λ □ (0,1), trading off growth rate for reduced variance.
22. For power utility $U(W) = W^\alpha / \alpha$ with α < 1, the optimal fraction is f* · α / (1−α) or related expression.
23. The security level approach sets a minimum wealth floor W_{min} and then applies Kelly on the surplus / excess wealth above that floor.
24. Hakansson (1971) showed the Kelly criterion is equivalent to maximizing log / logarithmic utility for a specific class of investors.
25. As the risk-aversion parameter α → 1, the fractional Kelly strategy converges to the full Kelly fraction.

VI. Error Sensitivity & Simulations (Chs. 31–38)

26. Errors in estimating m (mean return) by δm shift the optimal fraction by approximately δm / σ^2.
27. Simulation studies in the book show that Kelly betting leads to ruin in the short run if the true edge is less than / below the estimated edge.
28. MacLean & Ziemba (1999) demonstrated that with parameter uncertainty, the fractional / conservative Kelly fraction is robust.
29. In simulations with 700 trials, the Kelly bettor beat the fixed-fraction bettor in over 95 % of scenarios.
30. Sensitivity analysis shows that a 10% overestimate of edge results in a growth-rate loss of approximately 1% / small amount.

VII. Applications — Thorp, Paradoxes & Real Markets (Chs. 39–50)

31. Edward Thorp's fund, Princeton-Newport Partners, achieved an annualized return of approximately 15–20 % from 1969–1988.
32. The St. Petersburg paradox is resolved by Kelly because log utility assigns decreasing / diminishing marginal value to each additional dollar.
33. The 'Siegel paradox' shows that in foreign exchange, both parties can have positive expected log / geometric returns simultaneously.
34. MacLean, Sanegre, Zhao & Ziemba (2004) used Kelly with a security / floor constraint to guarantee capital growth.
35. Thorp observed that in practice, trading costs and market impact / slippage reduce the effective Kelly fraction below the theoretical optimum.

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